

Systematic biases in spatial working memory: a reanalysis of open data with pre-specified tests

Nikita Filippov. October 2026. Code: <https://github.com/stockman2/schizo-study-v1.git>

Summary. This document is a worked example of a reanalysis of open data. We reanalysed the data of Stein, Barbosa et al. (2020) on single-item spatial working memory. Every decision rule was fixed before the corresponding computation, every method was first run on synthetic data with a known answer, and a part of the data was reserved for confirmation. The main results reproduce known findings; we do not claim that they are new (see “Relation to prior work”). About half of the error variance is a systematic bias that depends on the target location. Its main component is an attraction toward the diagonals. The strength of this attraction is proportional to the random variance, as in the category adjustment model. A second group of biases grows at an approximately constant rate during the delay; the attraction toward the previous target belongs to this group. A model with a drift during the delay predicts the correct size of the location dependence of the random variance. A model with a single correction at the time of the response predicts an effect that is too large by a factor of about two. In patients with schizophrenia the biases of the second group are smaller; this statement is exploratory.

How to read this document. Each result is stated in the same form: the hypotheses that were compared, the statistic, the decision rule, the data, and the outcome. Every decision rule was written in the analysis script before the corresponding computation. Section 2 defines all quantities.

Relation to prior work

The phenomena analysed here are known. We list the correspondences.

- *Category adjustment model* (Huttenlocher, Hedges and Duncan, 1991). Estimates of the location of a dot in a circle are biased away from the horizontal and vertical axes, toward the centres of the quadrants. The estimate is a weighted mean of an inexact memory and the centre of the category. The weight of the category increases with the uncertainty, for example with the delay. Results 1 and 2 reproduce this. Our hypothesis (H1) is the weighting rule of this model, and our model R is a model of this type.
- *Dynamic field theory* (Spencer and Hund, 2002, 2003; Schutte and Spencer, 2002, 2009). In this neural field model the memory drifts during the delay. This literature separates geometric biases away from reference axes, drift during the delay, and biases toward previously used locations. Our classes U and D (Section 4) are close to this separation. Result 5 is close to the drift toward previous locations reported there.
- *Attractor dynamics with drift and diffusion* (Panichello et al., 2019). Errors in colour memory were described by diffusion together with a drift toward attractor states. The attraction is stronger when the noise is larger. Our model D is of this type.
- *Schizophrenia*. Badcock et al. (2008) reported that imprecise spatial memory in patients is accompanied by a larger bias from category-level representations. Stein et al. (2020) reported that the attraction toward the previous target is absent in patients. In the present data the category bias is not larger in patients (Result 8).
- *Efficient coding* (Wei and Stocker, 2015) is the alternative hypothesis in Result 10.

What this document adds. (i) A procedure: pre-specified rules, validation on synthetic data, and a held-out part of the data. (ii) A quantitative test of the weighting rule over delays within participants, with a replication. (iii) Relation (5): a gradual attraction and a single correction of the same size differ by a factor of two in their effect on the local variance. This gives a test between the two types of model. We did not find this test in the works that we read; our search was not exhaustive. (iv) An exploratory comparison of the late drift between patients and controls.

1. Task and data

Task. A dot is shown for 0.25 s on a circle around the fixation point. After a delay $t \in \{0, 1, 3\}$ s the participant reports the location of the dot with the mouse. For $t = 0$ the dot remains visible until the response begins; hence there is no memory interval at $t = 0$.

Samples. *Baseline sessions:* 52 participants (19 healthy controls, 17 patients with schizophrenia, 16 patients with anti-NMDAR encephalitis), 52 394 trials after the exclusion criteria of the original study (this number

equals the published number). *Post sessions* (3–12 months later): 22 of these participants (8 controls, 14 patients with encephalitis), 23 356 trials.

Roles of the samples. A statement is called *exploratory* if it was found in the baseline sessions without a rule fixed in advance. A statement is called *confirmed* if it satisfied a rule fixed in advance on data that had not been used to formulate it. The post sessions were reserved for confirmation. Each quantity was computed in the post sessions only once.

2. Definitions

Angles are measured in degrees in the coordinates of the data file. The *cardinal axes* are the directions 0° , 90° , 180° , 270° . The *diagonals* are the directions 45° , 135° , 225° , 315° (Fig. 1a).

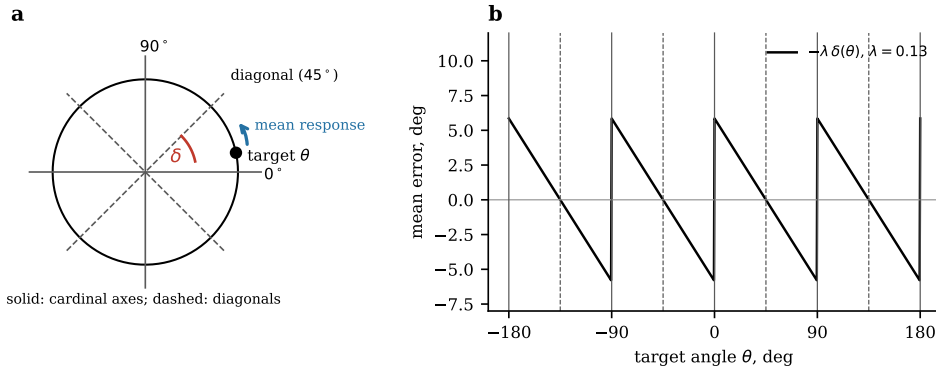


Figure 1. (a) Geometry. The signed distance δ from the target to the nearest diagonal; an attraction toward the diagonal moves the mean response as shown. (b) The mean error $-\lambda\delta(\theta)$ produced by an attraction with coefficient λ .

For a trial with target angle θ and response angle r , the *error* is $\varepsilon = r - \theta$ (wrapped to $[-180^\circ, 180^\circ]$). Let d be the angle from the current target to the target of the previous trial. All quantities below are computed for each participant and each delay t separately, after the mean error is subtracted.

- *Serial bias* β . We fit $\varepsilon = \beta G(d)$ by least squares, where G is the first derivative of a Gaussian with width 0.8 rad, normalised to maximum 1 (the function used in the original study). $\beta > 0$ means that the response is attracted toward the previous target. The term $\beta G(d)$ is then subtracted.
- *Bias map and random variance*. The bias map is the mean error as a function of θ . Its variance V_{map} is estimated without bias from two disjoint halves of the trials (24 bins of 15° ; the product of the two half-means is averaged over bins). The *random variance* is $V = \text{Var } \varepsilon - V_{\text{map}}$.
- *Attraction coefficient* λ . Let $\delta(\theta) \in [-45^\circ, 45^\circ]$ be the signed distance from θ to the nearest diagonal. We fit $\varepsilon = -\lambda\delta(\theta) + a_1 \sin \theta + b_1 \cos \theta + a_2 \sin 2\theta + b_2 \cos 2\theta + c$. $\lambda = 0.13$ means that the mean response moves 13% of the distance from the target to the diagonal (Fig. 1b).
- *Shift in one direction*. The first harmonic $a_1 \sin \theta + b_1 \cos \theta = A_1 \sin(\varphi_1 - \theta)$ corresponds to a displacement of all remembered locations toward the direction φ_1 with amplitude A_1 .
- *Attraction toward the axis* 0° – 180° . The second harmonic equals $A_2 \sin 2(\varphi_2 - \theta)$. We define s_2 as its component for $\varphi_2 = 0^\circ$; $s_2 > 0$ means attraction toward the axis 0° – 180° .

Group statistics. A group value is the mean over participants. Intervals are 95% bootstrap intervals (2000 resamples of participants) unless stated otherwise. A prediction that is computed from group values is recomputed in every resample.

Compatibility rule. Several tests compare an observed value with a prediction. We call a hypothesis *compatible* with the data if the interval of the difference (observed minus predicted) contains zero. A hypothesis is *accepted* if it is the only compatible hypothesis among those compared.

Validation of methods. Before each computation on the data, the script was run on synthetic data that were generated under each of the compared hypotheses. A script was used only after it had returned the correct answer for every synthetic data set.

3. Results on the data

Result 1. *About half of the error variance is systematic.*

Statistic. The share of V_{map} in the growth of the total variance between $t = 1$ and $t = 3$ s.

Rule. Share ≤ 0.2 : the growth is random. Share ≥ 0.5 : the growth is systematic. Otherwise: both.

Data. Baseline sessions.

Outcome. In controls the total variance is 14.5, 38.8, 57.2 deg² at $t = 0, 1, 3$ s, and V_{map} is 4.7, 16.7, 27.0 deg². The share is 0.56 [0.42, 0.67] in controls, 0.52 in patients with schizophrenia, 0.57 in patients with encephalitis. The maps of different participants are correlated ($r \approx 0.5$ –0.6). Harmonics 4, 8, 12 contain 61% of V_{map} : the responses are attracted toward the diagonals.

Result 2. *The attraction toward the diagonals is proportional to the random variance.*

Hypotheses. (H1) The attraction follows the uncertainty (the weighting rule of the category adjustment model):

$$\lambda(t)(1 - \lambda(t)) = V(t)/V_p. \quad (1)$$

(H2) The trace moves toward the diagonal at a constant relative rate:

$$1 - \lambda(t) = (1 - \lambda(0))e^{-kt}. \quad (2)$$

Statistic. The observed $\lambda(3)$ and two predictions computed from $t = 0$ and $t = 1$. Under (H1): $\lambda(3)(1 - \lambda(3)) = \lambda(1)(1 - \lambda(1)) V(3)/V(1)$. Under (H2): $1 - \lambda(3) = (1 - \lambda(0))[(1 - \lambda(1))/(1 - \lambda(0))]^3$.

Rule. Compatibility rule (Section 2).

Data. Baseline sessions (first test); post sessions (replication).

Outcome. Table 1 and Fig. 2a. (H2) is not compatible and (H1) is compatible in both samples. A second test in the post sessions: the ratio $\lambda(1 - \lambda)/V$ is constant over the delays (Fig. 2b). The rule required that the intervals of the ratios of consecutive delays contain 1; they are 0.95 [0.80, 1.27] and 1.03 [0.88, 1.22]. The constant is $V_p \approx 310$ –320 deg².

Table 1. Attraction coefficient at $t = 3$ s: observation and two predictions.

	Observed	Prediction (H2)	Prediction (H1)
Baseline, $n = 52$	0.132 [0.117, 0.147]	0.192 [0.169, 0.216]	0.127 [0.107, 0.149]
Post, $n = 22$	0.125 [0.102, 0.152]	0.198 [0.158, 0.237]	0.121 [0.089, 0.159]

Result 3. *A shift in one direction has a constant part and a part that follows the random variance.*

Hypotheses. For the growth of the shift between 1 and 3 s: (H1) the increment is proportional to the increment of V ; (H2) the shift grows at a constant rate.

Statistic. Let $p_1(t)$ be the component of the first harmonic in the direction -85° (this direction was found in the exploratory analysis and was then fixed). The observed increment $p_1(3) - p_1(1)$ is compared with $[p_1(1) - p_1(0)] [V(3) - V(1)]/[V(1) - V(0)]$ under (H1) and with $2[p_1(1) - p_1(0)]$ under (H2).

Rule. Compatibility rule. In addition, the direction is confirmed if its interval lies in $[-135^\circ, -45^\circ]$.

Data. Exploration: baseline sessions. Confirmation: post sessions.

Outcome. The direction is -89° $[-107^\circ, -70^\circ]$. The amplitude is 1.3° at $t = 0$ and 2.5° at $t = 3$ s. The observed increment is 0.23° . Under (H1) the prediction is 0.22° (difference 0.01° $[-0.54, 0.62]$). Under (H2) it is 1.87° (difference -1.64° $[-2.75, -0.59]$). (H1) is accepted.

Result 4. *The attraction toward the axis 0° – 180° is a late drift.*

Hypotheses. The mean of s_2 does not change during the first second and grows afterwards. This is the opposite of (H1): V grows mainly during the first second (Table 2).

Statistic. The early increment $F = s_2(1) - s_2(0)$ and the late rate $L = [s_2(3) - s_2(1)]/2$.

Rule. Confirmed if the interval of L lies above zero and the interval of $F - L$ lies below zero.

Data. Exploration: baseline sessions. Confirmation: post sessions.

Outcome. Post sessions: $L = 0.45^\circ/\text{s}$ [0.23, 0.67], $F = -0.13^\circ$ $[-0.70, 0.42]$, $F - L = -0.58^\circ$ $[-1.16, -0.01]$ (Fig. 2c). Exploration: $L = 0.52^\circ/\text{s}$ [0.39, 0.67]. The statement is confirmed; the upper bound of $F - L$ is close to zero.

Result 5. *The attraction toward the previous target grows faster than the random variance.*

Hypotheses. For the growth of β : (H1) the increment is proportional to the increment of V ; (H2) β grows at a constant rate.

Statistic. The observed $\beta(3)$ is compared with $\beta(0) + [\beta(1) - \beta(0)][V(3) - V(0)]/[V(1) - V(0)]$ under (H1) and with $\beta(0) + 3[\beta(1) - \beta(0)]$ under (H2).

Rule. Confirmed if the interval of the difference between the observation and the prediction of (H1) lies above zero.

Data. Post sessions (β had not been computed there before).

Outcome. $\beta = -0.29^\circ, +0.34^\circ, +2.09^\circ$ at $t = 0, 1, 3$ s. Under (H1) the prediction is 0.65° ; the difference is $+1.44^\circ$ [0.80, 2.05]. Under (H2) the prediction is 1.59° ; the difference is $+0.51^\circ$ [-0.63, 1.57]. The statement is confirmed, and (H2) is compatible.

Result 6. *Individual bias fields are stable.*

Statistic. The correlation, across participants, between the pre and post sessions of the individual components of the first harmonic and of the second harmonic (the group mean is subtracted), $t = 1$ s.

Rule. Confirmed if the lower bound of the interval is above zero.

Outcome. $r = 0.52$ [0.15, 0.77] for the first harmonic and $r = 0.69$ [0.51, 0.87] for the second harmonic.

Result 7. *The random variance is larger near the cardinal axes than near the diagonals, and the difference grows with the delay.*

Statistic. The ratio of the residual variance for targets within 15° of an axis to the residual variance for targets within 15° of a diagonal.

Rule. Confirmed if the interval of the difference of the ratio between $t = 3$ and $t = 0$ s lies above zero.

Data. Exploration: baseline sessions (ratio 1.15, 1.37, 1.69 at $t = 0, 1, 3$ s). Confirmation: post sessions.

Outcome. The difference is 0.35 [0.05, 0.71]. In these ratios the residual was computed after the sawtooth $-\lambda\delta$ was removed. If a flexible map (harmonics 1–12) is removed instead, the logarithm of the ratio is 0.24 [0.17, 0.32] at 1 s and 0.37 [0.25, 0.48] at 3 s (baseline sessions).

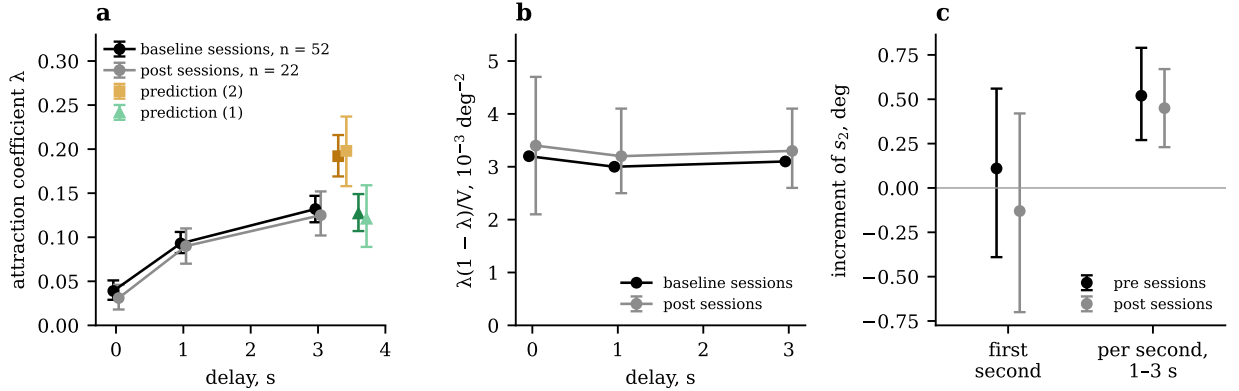


Figure 2. (a) Attraction coefficient λ and the two predictions for $t = 3$ s (prediction (1) is (H1), prediction (2) is (H2)). (b) The ratio in (1) at the three delays; intervals were not computed for the baseline sessions. (c) Increment of s_2 in the same 22 participants in two sessions.

4. Two classes of biases

Results 2–5 separate the biases into two classes (Table 2).

Class U (the bias follows the uncertainty): the attraction toward the diagonals and the shift in one direction. Their increments are proportional to the increments of V . Most of the growth occurs during the first second.

Class D (drift): the attraction toward the axis 0° – 180° and the attraction toward the previous target. Their growth between 1 and 3 s is larger than the growth predicted from V .

This distinction is close to the distinction between geometric biases and delay- or experience-dependent biases in the literature on the dynamic field theory.

Table 2. Growth of each component in the post sessions ($n = 22$). The unit is deg^2 for V and degrees for the other components; λ is dimensionless.

Component	Increment, $0 \rightarrow 1$ s	Increment, $1 \rightarrow 3$ s	Class
Random variance V	15.9	8.0	—
Attraction toward diagonals, λ	0.059	0.035	U
Shift in one direction, p_1	0.94	0.23	U
Attraction toward axis, s_2	-0.13	0.90	D
Attraction toward previous target, β	0.62	1.76	D

Result 8. *In patients with schizophrenia the class D biases are smaller (exploratory).*

Hypotheses. The late drift toward the axis is smaller in patients with schizophrenia than in controls. (The absence of the attraction toward the previous target in these patients is the published result of the original study.)

Statistic. The late rate L (Result 4) of each participant; the difference of the group means, controls minus patients.

Rule. Confirmed if the interval of the difference lies above zero.

Data. Baseline sessions. These data had been inspected before the rule was written. No independent sample of patients was available.

Outcome. $L = 0.60^\circ/\text{s}$ [0.49, 0.72] in controls and $0.34^\circ/\text{s}$ [0.09, 0.58] in patients; the difference is $0.26^\circ/\text{s}$ [0.01, 0.55]. The rule is satisfied; the lower bound is close to zero. For comparison, the late increment of β is $+2.04^\circ$ [1.44, 2.76] in controls and $+0.11^\circ$ [-0.71, 0.86] in patients. Class U is not reduced: V_{map} at 3 s is 27.0, 26.3, 24.3 deg^2 in the three groups.

We state the following **hypothesis** for future tests: *in schizophrenia the biases of class D are reduced and the biases of class U are preserved.* Badcock et al. (2008) reported a larger category bias in patients; the present data do not show this. A further test was not informative: the correlation, across participants, between the late increment of β and L is 0.02 [-0.28, 0.30], but the split-half reliabilities of these two individual quantities are 0.27 and -0.12.

5. A differential model for class U

Model D. Let x be the location of the memory trace. We assume

$$dx = -\frac{D(t)}{V_*} g(x) dt + \sqrt{2D(t)} dW, \quad (3)$$

where g is the shape of the attraction ($g = 0$ on the diagonals and on the axes), $D(t)$ is the noise intensity, and W is a Wiener process. The drift is proportional to the noise intensity. Models with diffusion and a drift toward attractor states were fitted to colour memory by Panichello et al. (2019). Let $\tau(t) = \int_0^t D(s) ds$. For one well with $g(x) = x$,

$$1 - \lambda = e^{-\tau/V_*}, \quad V = V_* \lambda (2 - \lambda). \quad (4)$$

For small λ this coincides with (1) with $V_p = 2V_*$. For $\lambda \leq 0.13$ the two forms cannot be distinguished in the data. The group means give $V_* \approx 150 \text{ deg}^2$, and $D \approx 9.5 \text{ deg}^2/\text{s}$ during the first second and $D \approx 2.8 \text{ deg}^2/\text{s}$ afterwards.

Model R (a model of the category adjustment type). The trace diffuses without drift and has variance V_{mem} at the end of the delay. The response is $m - \lambda g(m)$, where m is the final location of the trace and $\lambda = V_{\text{mem}}/(V_{\text{mem}} + V_p)$.

Difference between the models. Let $s(x) = g'(x)$. To first order in λ ,

$$V_D(x) \approx 2\tau(1 - \lambda s(x)), \quad V_R(x) \approx 2\tau(1 - 2\lambda s(x)). \quad (5)$$

A gradual attraction changes the local variance half as much as a single correction of the same size. Both models reproduce Result 2. They differ in the dependence of the random variance on the location.

Result 9. *Model R is rejected. Model D predicts the correct size of the effect but is not accepted.*

Hypotheses. Model D against model R.

Statistic. The residual variance in three bands of $|\delta|$: 0–15° (near the diagonal), 15–30° (middle), 30–45° (near the axis), and the ratios of the second and third to the first.

Calibration. For each participant the well parameter (V_* or V_p) was chosen so that the model reproduces the sum of λ over the delays, and the amount of noise at each delay was chosen so that the model reproduces V . Model trials were generated at the target angles of the real trials and were analysed by the same code. The distribution of the variance over the circle was not used in the calibration.

Rule. A model is compatible if the 99% interval of the difference between data and model contains zero for every ratio at $t = 1$ and $t = 3$ s. A model is accepted if it is the only compatible model.

Data. Baseline sessions (main test); post sessions (for reference).

Two rounds. In the first round g was a sawtooth, $g(x) = \delta(x)$, and only the ratio “near axis / near diagonal” was tested. Model D failed at $t = 3$ s. In the second round g was a smooth shape (harmonics 4, 8, 12) fitted so that the mean bias map of the model equals the measured map, and both ratios were tested. The second round is a revision after a failed test on data that had been inspected.

Outcome. Table 3 and Fig. 3. Model R overestimates the ratio near the axes by 1.1–1.6 in all eight comparisons; this agrees with (5). Model D with the sawtooth gives the correct level near the axes, but the middle band does not differ from the band near the diagonal (model 1.00–1.02; data 1.19 and 1.26). Model D with the smooth shape reproduces the middle band (1.14 and 1.16), but the ratio near the axes is too large at 1 s and in the post sessions. The data lie between the two variants of model D. By the rule, neither model is accepted in either round.

Table 3. Ratio of the residual variances, near axis / near diagonal. An asterisk indicates that the 99% interval of the difference between data and model does not contain zero.

Sample	Delay	Data	D, sawtooth	D, smooth	R, sawtooth	R, smooth
Baseline	1 s	1.37 [1.25, 1.50]	1.32	1.60*	2.45*	2.48*
	3 s	1.69 [1.51, 1.88]	1.44*	1.78	2.81*	2.90*
Post	1 s	1.22 [1.08, 1.37]	1.35	1.59*	2.52*	2.44*
	3 s	1.25 [1.02, 1.54]	1.43	1.81*	2.81*	2.67*

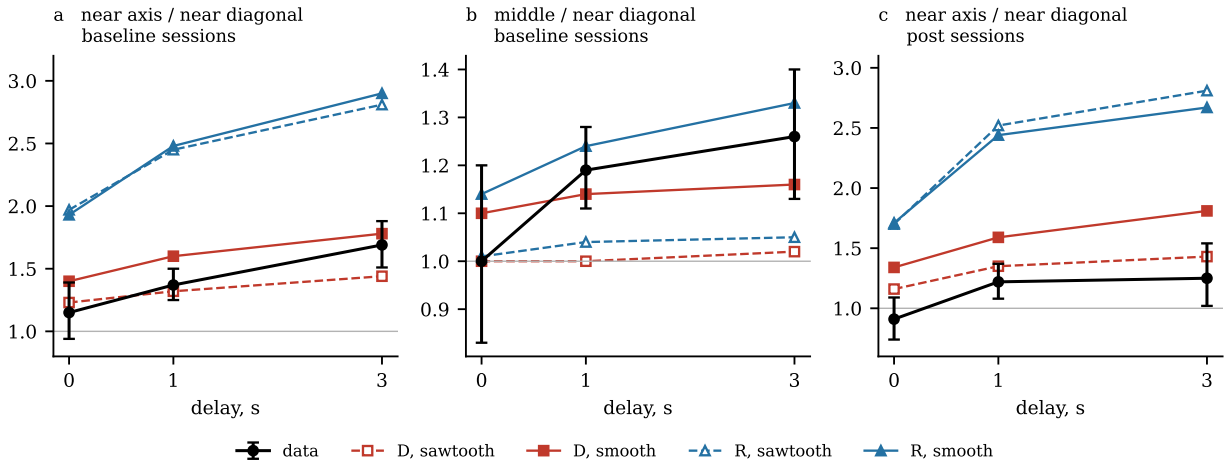


Figure 3. Ratios of residual variances: data (95% intervals) and four model variants. The delay 0 s is shown for reference.

6. Scope: perception of orientation and motion direction

Result 10. *The potential-well picture holds for memory of location and does not hold for the perception of orientation.*

Hypotheses. In a potential well the responses are attracted toward locations where the random variance is small. An alternative (efficient coding) predicts that the responses move away from values that are encoded precisely, that is, toward locations where the random variance is large.

Statistic. For each participant: λ (sign: positive toward the diagonals or oblique orientations) and $m = \ln(\text{variance near the axes}/\text{variance near the diagonals})$, computed after a flexible map is removed.

Rule. For each experiment: “well” if the intervals of the mean λ and of the mean m exclude zero and the signs are equal; “opposite” if both exclude zero and the signs differ; otherwise undecided. For a stimulus

type: the answer is the verdict of at least two thirds of the decided experiments, if at least three are decided. *Data.* 30 open experiments (orientation or motion direction) from the collection of Ozkirli, Chetverikov and Pascucci (2025); the data of Section 1 for comparison.

Outcome. Location memory (Section 1 data): “well” at both delays ($m = 0.24$ and 0.37). Orientation (22 experiments): $\lambda > 0$ in 19 experiments (0.07–0.20, the same range as in Result 2), but 2 experiments are “well”, 5 are “opposite”, and 15 are undecided; the verdict by the rule is “opposite”. Motion direction (8 experiments): 2 are “well” and none is “opposite”; fewer than three are decided. The two “well” experiments are working memory experiments.

7. Statements that were not confirmed

- *Alignment of individual axes.* In the baseline sessions the individual axis of the second harmonic lies within 22.5° of 0° or 90° in 38 of 52 participants. The rule for the post sessions required at least 16 of 22; the observed number is 13.
- *Recovery from encephalitis.* We predicted that $\lambda(1 - \lambda)/V$ increases from the pre to the post session in patients. The change is $+0.0005$ $[-0.0003, 0.0011]$; the interval contains zero.
- *Differences between participants.* If (1) held across participants with a common V_p , then λ and V would be correlated. The rank correlation at $t = 1$ s is 0.14 $[-0.15, 0.41]$ (reliability of both quantities: 0.96). Relation (1) holds within a participant over time.
- *Growth law of $V(t)$.* At $t = 0$ there is no memory interval, and only two positive delays are available. Diffusion with an initial loss cannot be distinguished from saturation.

8. Limitations

- Three delays give two time intervals.
- The data do not distinguish an attraction that occurs during the delay from an attraction at the time of the response whose size increases with the delay; only the two specific variants of model R were rejected.
- The post sessions contain the same participants as the baseline sessions and no patients with schizophrenia. All statements about schizophrenia rest on the baseline sessions.
- The orientation of the screen coordinates is not known to us. Gaze was not recorded. The patients were medicated.
- Model D does not contain the class D biases, lapses, or trial-to-trial changes of precision.

9. Predictions for data with longer delays

1. In model D the random variance saturates at V_* (125–155 deg^2). In model R the observed variance does not exceed $V_p/4$ and then decreases.
2. If the class D biases grow at a constant rate, then at delays of 8–20 s they exceed the class U biases.
3. In patients with schizophrenia the class D biases are reduced at all delays, and the class U biases are not.

Use of AI

The analysis scripts and the text of this document were prepared with the assistance of a large language model (Claude, Anthropic). Nikita Filippov ran all analyses, checked the results, and is responsible for the content.

Sources

Stein H., Barbosa J., et al. Reduced serial dependence suggests deficits in synaptic potentiation in anti-NMDAR encephalitis and schizophrenia. *Nat. Commun.* 11, 4250 (2020). Data: github.com/comptelab/serialNMDA.

Ozkirli A., Chetverikov A., Pascucci D. Large-scale mega-analysis indicates that serial dependence deteriorates perceptual decision-making. *Nat. Hum. Behav.* (2025), and the original studies listed in its data repository.

- Huttenlocher J., Hedges L. V., Duncan S. Categories and particulars: prototype effects in estimating spatial location. *Psychol. Rev.* 98, 352–376 (1991).
- Spencer J. P., Hund A. M. Prototypes and particulars: geometric and experience-dependent spatial categories. *J. Exp. Psychol. Gen.* 131, 16–37 (2002).
- Spencer J. P., Hund A. M. Developmental continuity in the processes that underlie spatial recall. *Cogn. Psychol.* (2003).
- Schutte A. R., Spencer J. P. Generalizing the dynamic field theory of the A-not-B error beyond infancy. *Child Dev.* 73, 377–404 (2002).
- Schutte A. R., Spencer J. P. Tests of the dynamic field theory and the spatial precision hypothesis. *J. Exp. Psychol. Hum. Percept. Perform.* (2009).
- Panichello M. F., DePasquale B., Pillow J. W., Buschman T. J. Error-correcting dynamics in visual working memory. *Nat. Commun.* 10, 3366 (2019).
- Badcock J. C., et al. Examining encoding imprecision in spatial working memory in schizophrenia. *Schizophr. Res.* (2008).
- Wei X.-X., Stocker A. A. A Bayesian observer model constrained by efficient coding can explain ‘anti-Bayesian’ percepts. *Nat. Neurosci.* 18, 1509–1517 (2015).
- Gold J. M., et al. Refining the empirical constraints on computational models of spatial working memory in schizophrenia. *Biol. Psychiatry Cogn. Neurosci. Neuroimaging* 5, 913–922 (2020).
- Starc M., Murray J. D., et al. Schizophrenia is associated with a pattern of spatial working memory deficits consistent with cortical disinhibition. *Schizophr. Res.* 181, 107–116 (2017).